Enhancing the oxidative cleavage of biomass derived vicinal diols with the use of hierarchical Fe-ZSM-5 zeolites

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Introduction: The ongoing effects of climate change, driven by greenhouse gas emissions and the depletion of fossil resources are making more intense the search for alternative renewable sources, such as biomass. Vicinal diols derived either from lignocellulosic biomass or fatty acids can be converted through oxidative cleavage reactions into various mono- and dicarboxylic acids, which are important polymer precursors. Currently, this reaction is typically conducted via ozonolysis, a high cost and energy demanding process. A more sustainable approach involves using hydrogen peroxide (H_2O_2), a "green" oxidant, in combination with heterogeneous catalysts, offering a promising solution for the oxidative cleavage of vicinal diols. In our previous work it was demonstrated that monomeric iron species on the surface of microporous ZSM-5 were highly active and selective for small vicinal diol cleavage. However, a decrease of reaction rates was observed when larger molecules were used [1]. Consequently, this study explores the use of hierarchical Fe-ZSM-5 catalysts, containing a network of micro- and mesopores, that can improve the accessibility to the active catalytic sites.

Results and discussion: For this work, two hierarchical ZSM-5 catalysts were prepared. The one catalyst was synthesized via a top-down method according to which the convetional ZSM-5 zeolite was subjected to desilication with the use of basic solution and was denoted as Fe-ZSM-5_DS. The second catalyst was prepared via a bottom-up approach, by using a carbon material as hard template for the formation of mesopores (Fe-ZSM-5_CT). The presence of mesoporosity was confirmed through N_2-physisorption and TEM. The carbon-templated Fe-ZSM-5_CT showed a spongy mesoporous structure, while the desilicated Fe-ZSM-5_DS exhibited a hollow structure. For both ZSM-5 and hierarchical materials iron was introduced via ion-exchange in 0.4-1.4 wt.% loading (ICP-OES). The monomeric nature of iron species on the modified zeolite catalysts was confirmed with XAS

and UV-Vis measurments. The catalytic results revealed that under mild reaction conditions (25 °C, in water), the introduction of mesoporosity increased the reaction rates, with Fe-ZSM-5_DS presenting the best overall activity. This outcome was due to the higher Fe-exchange capacity that could be achieved on the desilicated sample (1.4 wt.%), compared to the other two samples (~0.4 wt.%). Regarding the activity of the average Fe-species, Fe-ZSM-5_CT showed the highest TOF for all terminal vicinal diols as well as bio-derived polyols (**Figure 1**), attributed to the larger mesopores (~20 nm), compared to the desilicated sample Fe-ZSM-5_0.3DS (~10 nm) [2].

Conclusion: The additional mesoporosity in both carbon templated and desilicated Fe-ZSM-5 cataysts enhanced the oxidative cleavage of large vicinal diols, surpassing the performance of the conventional microporous Fe-ZSM-5. For Fe-ZSM-5_DS, the higher catalytic activity could be attributed not only to the presence of mesopores but also to the higher Fe loading, compared to Fe-ZSM-5. In the case of Fe-ZSM-5_CT, despite the same amount of monomeric Fe-species with the Fe-ZSM-5, the presence of mesopores facilitated mass transport, ensuring better accessibility to the active sites.

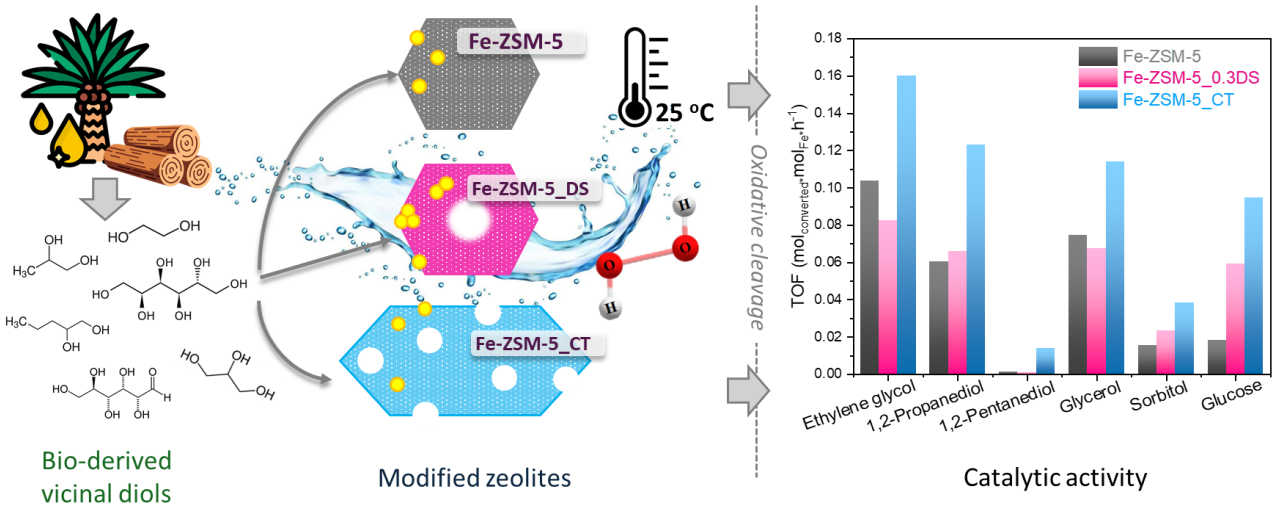

Figure 1. Oxidative cleavage of biomass derived vicinal diols in presence of Fe loaded hierarchical ZSM-5 zeolites.

References

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